A. Electronic Measurements & Instrumentation – EE3300

EMI 1 – Introduction to Electronic Measurements

OBJECTIVES

1. To understand the basic operation of the function generator, oscilloscope and multimeter.

2. To use the oscilloscope to observe signals created by various sources.

EQUIPMENT

Function generator; Oscilloscope; Multimeter; Step-down transformer; Microphone;

4 Coaxial cables with BNC connectors;

1 T connector;

1 BNC-banana adapter

PROCEDURE

1. Familiarize yourself with the main controls on the function generator: the power switch, the function frequency selector, the function frequency multiplier, the offset knob, the amplitude knob, and the output function shape. Using the controls, set the generator to create a sine wave with a frequency of 3 kHz.

2. Next, familiarize yourself with the basic settings on the oscilloscope: the power button, the vertical scale knob, and the horizontal scale knob. Ignore the other buttons for now.

3. Connect one of the oscilloscope probes to the output wire for the function generator. Turn both instruments on. Adjust the vertical and horizontal knobs on the scope to get a clear view of the signal. *Sketch the diagram*.

4. Adjust the knobs on the function generator to see what different signals (square, triangular) look like. *Also sketch the diagrams*.

(a) Specifically, what are the minimum and maximum signal amplitudes that can be created by your function generator?

(b) What are the minimum and maximum signal frequencies it can produce?

5. Now that you have a basic understanding of how to work the oscilloscope, we can do something interesting. Let's look at the electrical signal that comes from a

standard wall socket. To prevent you from being electrocuted, we will use a transformer to step down the voltage. You can think of this as a box which reduces

the voltage (in this case by a factor of 10). Connect the scope probe to the transformer output and then plug the transformer into the wall.

(a) *Sketch the signal*.

(b) What is its amplitude and frequency?

6. Let's look at another type of signal on the oscilloscope: sound waves. Unplug the scope probe from the function generator, and plug it instead into the microphone on your bench. The microphone converts sound waves in the air into electrical signals that can be analyzed on the scope.

7. Sing a note into the microphone and adjust the scope so that you can clearly see the electrical signal. Now, talk into the microphone and see what the signal looks like. Say a particular vowel, let's say the `e' in hello into the microphone.

(a) *Sketch the resulting signal and use the scope to figure out the frequency*.

(b) Compare the frequency of your voice to that of others in the lab. What is the range of frequencies you find?

(c) Scream a high pitch into the microphone. *Sketch the shape of the signal produced*. How does it differ from the low pitch signal?

8. Using a BNC-T connector, connect the output of the function generator (FG) to the multimeter while leaving the oscilloscope connected. For the multimeter connection, use a coaxial cable with a BNC plug on one end and a BNC-to-banana plug adapter on the other end. The positive lead of the banana plug adapter should be in the top right jack of the multimeter and the negative in the middle right jack. In order to tell which side of the banana plug is negative, look for a small ridge on one side. This side is the ground side. On some plugs, the letters GND are even written on this ridge. This ridge is called an ear, leading to the “ear to the ground” rule. Among other things, the multimeter is useful for measuring **root-mean-square** (RMS) voltage. The RMS value is often defined as:

$V_{rms} = 0.707\ V_p$ where V_p is the peak voltage (1)

Turn on the multimeter and measure the RMS voltage. Change the amplitude of the FG to 5 V_{pp} (peak-to-peak) and measure the corresponding RMS voltage.

9. Now investigate an AC signal with a DC offset. Change the FG back to the 2-V sine wave. Press the *Sine Wave* button on the top row and hold it. The FG is now set to output only a DC voltage. Adjust the output voltage to 1 V by turning the dial. The oscilloscope should now display a DC voltage at +1 V. Press the sine wave key. The

FG is now producing a 1 kHz sine wave with a 1 V DC offset. Check to make sure the o-scope coupling is DC. Notice the ground indicator to the right of the oscilloscope screen showing the 0 V level. Measure the RMS voltage using both the multimeter and the oscilloscope. Change the channel 1 coupling to AC and observe what happens to the oscilloscope's RMS reading. What does the AC coupling do? Repeat the RMS measurements for a 10 V_{pp} sine wave with and without a 5 V DC offset. Remove the offset by reversing the procedure for changing the DC offset.

10. Change the output waveform to a square wave (10 V_{pp}, no DC offset) by pressing the square wave key in the top row. Measure the RMS voltage with the multimeter and the peak-to-peak voltage with the oscilloscope. (Make sure % duty = 50 on the FG.)

11. Repeat the RMS voltage measurement with a triangle wave (10 V_{pp}, no DC offset). Again, record all readings.

DISCUSSION QUESTIONS

1. Explain the purpose of the following oscilloscope feature: AC, DC, GND coupling of the signal.

2. Describe what rms voltage means. For a sine wave, show that V_{rms} is equal to the following (work out the integral in your lab notebook):

3. When there is a DC offset in a signal, why does the multimeter give a different value for the rms voltage than the scope?

4. Explain the purpose of the following oscilloscope features: Auto level, Auto, Normal and Single trigger modes, AC/DC trigger coupling.

5. What scope settings (trigger, V/div, and time/div) would you use to capture a transient (single shot) signal consisting of two 1 Volt pulses, 100 ns in duration and separated by 10 ms? First sketch out what you'd like the scope screen to look like, then determine the settings to achieve t